

Can Natural Image Autoencoders Compactly Tokenize fMRI Volumes for Long-Range Dynamics Modeling?

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Abstract

Modeling long-range spatiotemporal dynamics in functional Magnetic Resonance Imaging (fMRI) remains a key challenge due to the high dimensionality of the four-dimensional signals. Prior voxel-based models, although demonstrating excellent performance and interpretation capabilities, are constrained by prohibitive memory demands and thus can only capture limited temporal windows. To address this, we propose TABLET (Two-dimensionally Autoencoded Brain Latent Transformer), a novel approach that tokenizes fMRI volumes using a pre-trained 2D natural image autoencoder. Each 3D fMRI volume is compressed into a compact set of continuous tokens, enabling long-sequence modeling with a simple Transformer encoder with limited VRAM. Across large-scale benchmarks including the UK-Biobank (UKB), Human Connectome Project (HCP), and ADHD-200 datasets, TABLET outperforms existing models in multiple tasks, while demonstrating substantial gains in computational and memory efficiency over the state-of-the-art voxel-based method given the same input. Furthermore, we develop a self-supervised masked token modeling approach to pre-train TABLET, which improves the model's performance for various downstream tasks. Our findings suggest a promising approach for scalable and interpretable spatiotemporal modeling of brain activity. Our code is available at <https://github.com/beotborry/TABLET>.

1. Introduction

The human brain is a spatiotemporal dynamic system whose activity can be non-invasively measured using functional magnetic resonance imaging (fMRI). A large body of work

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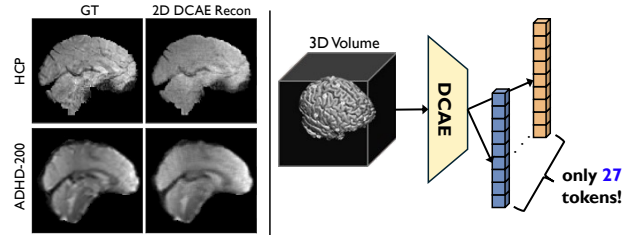


Figure 1. We show that a 2D natural image autoencoder can be transferred well to 4D fMRI data (left). Building on this, we propose to tokenize a single 3D volume of fMRI only into 27 tokens, enabling longer spatiotemporal modeling with a simple Transformer architecture (right).

has leveraged fMRI to investigate functional connectivity patterns for tasks such as neurological disorder diagnosis or demographic attribute prediction [5, 8, 15–18, 22]. Existing approaches can be broadly divided into two categories: ROI-based methods and voxel-based methods. ROI-based methods first define a set of regions of interest (ROIs) based on anatomical segmentation [23], extract their corresponding time-series signals, and then compute functional connectivity (FC) matrices as model inputs. Although this approach is computationally efficient for managing the high dimensionality of fMRI data, it has several limitations: performance strongly depends on the choice of ROIs, fine-grained 3D spatial structures may be lost, and aggressive compression can discard informative signals. To overcome these limitations, voxel-based methods such as TFF [18] and SwiFT [17] have been proposed. These methods directly process raw 4D fMRI data, thereby preserving spatial and temporal information, while also allowing detailed interpretation as they directly operate on the given image. However, due to the massive scale of fMRI volumes, the temporal length that could be simultaneously processed by the model is severely restricted (e.g., TFF and SwiFT use only 20 timesteps at once), potentially missing informative

long-range temporal dynamics, and limiting use for tasks that require longer-range interactions, such as the infraslow BOLD–LFP coupling and global arousal waves that unfold over tens of seconds [20, 25].

In this work, we aim to improve voxel-based fMRI modeling by *tokenizing* fMRI volumes into a compact set of continuous tokens, thereby enabling Transformers [35] to process substantially longer temporal sequences. To this end, we observed a strong perceptual information preservation capability of the Deep Compression Autoencoder (DCAE) [6] and aimed to leverage it, as it effectively tokenizes a 256×256 2D natural image into just 64 continuous tokens (a compression ratio of 32). Motivated by this, we ask whether a high-performing **2D** autoencoder trained on *natural images* can serve as an effective tokenizer for **4D fMRI** data.

Although it may seem counterintuitive, we demonstrate that such an autoencoder can effectively tokenize fMRI volumes while preserving both fine-grained and high-level functional information. Namely, by rearranging the tokens extracted from each 2D slice of a 3D fMRI volume, we compress an entire volume into *only* 27 continuous tokens, thereby dramatically reducing the input size and enabling substantially longer-sequence modeling with a simple Transformer encoder-based architecture even with limited VRAM. We dub our method **TABLET**, **T**wo-dimensionally **A**utoencoded **B**rain **L**atent **T**ransformer, which achieves consistent yet modest improvements compared to voxel-based baselines on demographic attribute prediction and attention-deficit hyperactivity disorder (ADHD) diagnosis tasks.

Our model drastically saves memory and computation costs compared to the most efficient voxel-based baseline under the same input data scale, allowing much longer sequences to be fed into the model. This unlocks the opportunity for future use cases that require such long sequences. Moreover, we demonstrate that self-supervised masked token modeling on fMRI data can effectively pre-train the model, thereby improving downstream task performance.

Our core contributions are summarized as follows:

- We propose a novel tokenization scheme that utilizes a 2D autoencoder, which is pre-trained on natural images. We then empirically demonstrate that, despite the apparent domain mismatch, it can serve as a practical general-purpose tokenizer for fMRI signals.
- Building on this tokenization, we construct a simple Transformer architecture to model substantially longer temporal sequences than prior state-of-the-art voxel-based models, leading to performance improvements across multiple tasks based on resting-state fMRI (rs-fMRI) datasets. This approach would allow application to tasks that require longer-range interactions.
- Furthermore, we show the feasibility of pre-training on

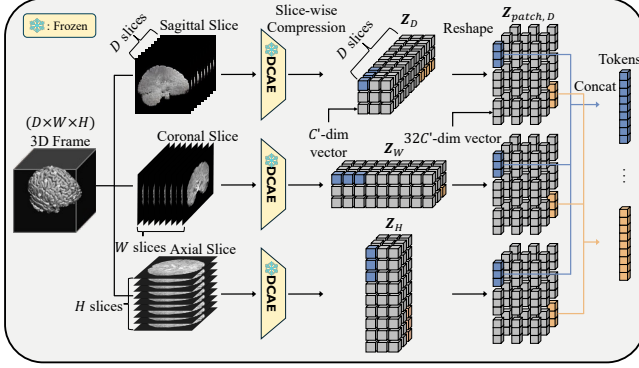
tokenized fMRI data using self-supervised masked token modeling, which can further improve downstream task performance.

2. Related Work

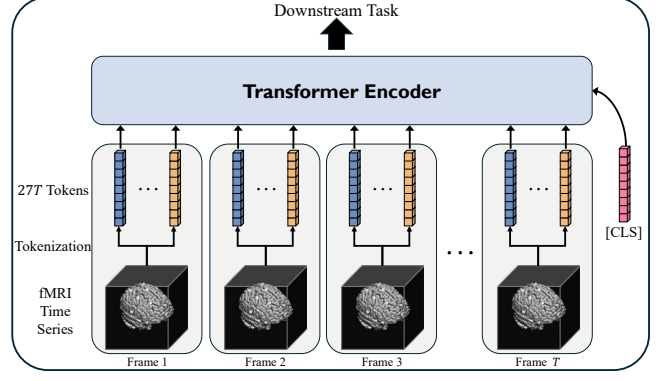
ROI-Based Methods. ROI-based methods parcellate the brain into ROIs and average the BOLD signals within each. The signals are transformed into FC matrices by computing the correlation between the time series of the ROIs. BrainNetCNN [16] treats the FC matrix as a 2D image and uses edge-to-edge, edge-to-node, and node-to-graph convolutional filters to utilize topological locality in ROI-based networks. Brain Network Transformer [15] adapts the Transformer architecture to process FC matrices as graphs of ROIs. meanMLP [22] is a lightweight MLP-based model that applies the same MLP repeatedly across parcellated fMRI time-series and averages the resulting embeddings across time before a final classification layer. Brain-JEPA [8] is a joint-embedding predictive architecture (JEPA) model pretrained on parcellated fMRI with spatiotemporal masking and gradient-based positioning. Even though computationally efficient, they are inherently limited by the strong pre-processing step that turns brain signals into FC matrices; it is heavily influenced by the choice of ROIs, and during the process, structural information as well as other signals can be discarded.

Voxel-Based Methods. Voxel-based methods process 4D fMRI volumes, enabling end-to-end learning of spatiotemporal features without ROI aggregation. TFF [18] operates on entire 4D volumes using a two-phase approach: self-supervised pretraining to reconstruct 3D volumes and fine-tuning. It captures fine-grained spatiotemporal dynamics, enabling transfer learning from unlabeled data. SwiFT [17] extends the Swin Transformer to 4D fMRI volumes with a 4D window multi-head self-attention mechanism and absolute positional embeddings. Voxel-based methods are free from the issues with ROI-based models; however, they are burdened with a higher memory and computation load, as they need to deal with high-dimensional data.

Self-Supervised Pretraining. Self-supervised learning (SSL) has emerged as a powerful pre-training framework for vision models, enabling scalable representation learning from unlabeled datasets. MAE [14] introduces an asymmetric encoder-decoder architecture, where a high portion of input image patches are masked. Visible patches are encoded by Vision Transformer (ViT), and the decoder reconstructs the masked patches in pixel space. MAE demonstrated superior downstream task transfer, such as classification and segmentation. SimMIM [37] proposes a masked image modeling (MIM) framework using hierarchical transformers and a simple linear prediction head. VideoMAE [34] extends the MAE framework to videos by randomly applying masks on spatio-temporal cubes across spatio-temporal di-



(a) Tokenization process of a 3D volume using a 2D encoder.



(b) Overview of TABLET.

Figure 2. In TABLET, each frame of the fMRI timeseries is tokenized by a 2D autoencoder, and the tokens are processed by a Transformer.

mensions. The model reconstructs the masked cubes, learning dynamics, and long-range interactions.

Deep Compression Autoencoder. Deep Compression Autoencoder (DCAE) [6] introduces an autoencoder framework for accelerating high-resolution diffusion models through extreme spatial compression ratios of up to $128\times$. DCAE achieves superior reconstruction quality at high compression levels by residual autoencoding. Residual autoencoding utilizes non-parametric shortcuts that enable the model to learn residuals. The encoder downsample blocks adapt a space-to-channel operation, and the decoder up-sample blocks use a channel-to-space operation. These non-parametric operations effectively preserve information without learned parameters.

3. Method

3.1. Tokenization of fMRI with 2D Natural Image Autoencoder

To enable voxel-based, long-range fMRI dynamics modeling, our goal was to design a tokenizer that could substantially compress fMRI voxels while minimizing information loss. We chose to employ the encoder part of an autoencoder, since it can handle tokenization while preserving coarse spatial topology. A straightforward strategy would be training an autoencoder directly on fMRI data. However, this approach is both computationally prohibitive and data hungry because training requires large sample sizes that are rarely available in medical imaging. Moreover, the resulting models often generalize poorly, as fMRI characteristics vary across scanners and acquisition protocols.

Therefore, we propose to directly use an autoencoder trained with natural images, without the need for further fine-tuning on medical data. Among existing options, we adopt DCAE, which achieves high compression rates while maintaining reconstruction fidelity.

To check if it would be feasible to use off-the-shelf autoencoders to tokenize fMRI data, we first measured how

much information would be preserved during tokenization. We compared an off-the-shelf 2D natural image DCAE (hereafter 2D DCAE) with a 3D DCAE trained directly on fMRI (hereafter 3D DCAE), as detailed in Sec. 4.4. We note that fMRI data are timeseries of 3D images, while the 2D DCAE only operates with 2D images. Therefore, we slice the data into 2D images and feed them into the autoencoder.

The 2D DCAE showed comparable information preservation capabilities across both fine-grained spatial details and global functional patterns. Based on this finding, we propose to tokenize each 3D volume independently using the 2D DCAE encoder, in a slice-wise manner, and apply this procedure across the entire fMRI sequence, as described below.

Tokenization of a 3D Volume with Slicing. Each fMRI frame is a 3D volume $\mathbf{X} \in \mathbb{R}^{1 \times D \times H \times W}$. The single channel is first duplicated across three channels to simulate an RGB structure, giving $\mathbf{X} \in \mathbb{R}^{3 \times D \times H \times W}$. One spatial dimension is then chosen as the slicing axis, so the volume becomes a stack of 2D images. For example, if we slice by the depth axis, the volume is treated as D images of shape $\mathbb{R}^{3 \times H \times W}$. Each image slice is compressed independently into a latent representation $\mathbf{Z} \in \mathbb{R}^{C' \times \frac{H}{32} \times \frac{W}{32}}$, where the factor of 32 is the DCAE’s spatial compression ratio.

Aggregation of 3 Axes. This procedure is repeated for all three slicing axes, producing three latent volumes: $\mathbf{Z}_D \in \mathbb{R}^{D \times C' \times \frac{H}{32} \times \frac{W}{32}}$, $\mathbf{Z}_H \in \mathbb{R}^{H \times C' \times \frac{D}{32} \times \frac{W}{32}}$, $\mathbf{Z}_W \in \mathbb{R}^{W \times C' \times \frac{D}{32} \times \frac{H}{32}}$. As we want to align the three possible latent volumes, the latents are grouped and concatenated along the uncompressed dimension (the slicing axis) by patches of size 32, reshaping them to $\mathbf{Z}_{\text{patch},D}, \mathbf{Z}_{\text{patch},H}, \mathbf{Z}_{\text{patch},W} \in \mathbb{R}^{32C' \times \frac{D}{32} \times \frac{H}{32} \times \frac{W}{32}}$. This yields $\frac{D}{32} \times \frac{H}{32} \times \frac{W}{32}$ tokens for each slicing variation, where each token corresponds to a position in the downsampled 3D grid and has a hidden dimension $32C'$. The tokens from the three variations are then concatenated with the tokens from other variations that belong to the same spatial posi-

tion, resulting in $\frac{D}{32} \times \frac{H}{32} \times \frac{W}{32}$ tokens per frame with hidden dimension $96C'$. In our case, $H = W = D = 96$ and $C' = 32$, which means each 3D volume of shape $(1, 96, 96, 96)$ is tokenized into 27 tokens with an embedding dimension of 3072. Finally, we note that tokenization is performed only once, and the tokens are cached for later use, making its computational cost negligible compared to the subsequent training process. We also investigate different aggregation schemes in Sec. C.3, which shows the downstream performance is not majorly affected by the specific aggregation scheme.

3.2. TABLET Model Architecture

To capture the spatiotemporal dynamics of tokenized fMRI sequences, we design a simple yet effective Transformer encoder [35], naming the pipeline **TABLET** (Two-dimensionally Autoencoded Brain Latent Transformer). The architecture is built on a standard Transformer encoder backbone and integrates several modern components commonly adopted in large language models [13, 24]. In particular, we adopt grouped query attention [1] to efficiently handle long sequences, along with the rotary positional encoding [30]. In addition, we employ `F.scaled_dot_product_attention` from PyTorch [21], which offers both speed and memory savings. Before being fed into the Transformer, fMRI tokens are normalized and projected into a lower-dimensional embedding space via a linear layer. A `[CLS]` token is prepended to the sequence, followed by an additional normalization step to enhance training stability. Unless stated otherwise, the model is composed of 12 Transformer layers with 14 attention heads and 2 key-value heads, processing sequences of tokens from 256 frames at once ($T = 256$). We randomly sampled 256 frames from the entire sequence of each subject at every training iteration, while for validation, we used all of the frames by partitioning the sequence and averaging the outputs across partitions, following Kim et al. [17].

3.3. Self-supervised Pre-training with Masked Token Modeling

Taking inspiration from masked image modeling (MIM) approaches like SimMIM [37], we propose a masked *token* modeling (MTM) approach to pre-train the Transformer encoder of TABLET. The key difference is that instead of masking image pixels or patches, we choose to directly mask the tokenized representations of the image; starting from the tokens created by the 2D DCAE, we randomly mask some of the tokens by replacing them with `[MASK]` tokens. From the partially masked input tokens, we task the Transformer encoder to predict the masked tokens by passing the output tokens through a linear prediction head that reconstructs the input tokens. The model is trained through an $\mathcal{L}_1$ loss exclusively on the masked tokens:

$$L = \frac{1}{\Omega(\mathbf{Z}_M)} \|\mathbf{y}_M - \mathbf{Z}_M\|_1, \quad (1)$$

in which $\mathbf{Z}, \mathbf{y} \in \mathbb{R}^{96C' \times \frac{D}{32} \times \frac{H}{32} \times \frac{W}{32}}$ are the input tokens and the predicted tokens, respectively. The subscript M denotes the set of masked tokens, and Ω counts the number of elements (thus the number of masked tokens). We used a masking ratio of 0.5 as the default in our experiments.

Even though it would be possible to use a typical MIM approach by masking the brain volume directly, we chose MTM as it streamlines the pre-training process by removing the need to include the image encoder and decoder.

Masking Strategy. We mask the input tokens with a learnable mask token. Also, instead of masking the tokens in a completely random manner, the same masking pattern from a single frame is repeated across different frames, similar to the tube masking strategy found within VideoMAE [34]. This is a measure to prevent the model from “cheating” by looking at unmasked tokens in the same location from different frames.

4. Experimental Results

4.1. Experimental Setting

Datasets. We used resting-state fMRI data from 8,178 middle-aged and older adults from UK-Biobank (UKB) [31], from 1,061 healthy young adults in the Human Connectome Project (HCP) [29], and from 533 children and adolescents, including both individuals diagnosed with ADHD and healthy controls, included in ADHD-200 [4].

For UKB and HCP, we used the preprocessed data provided by UK-Biobank [3, 19] and HCP [29], which go through the preprocessing pipeline including bias field reduction, skull-stripping, cross-modality registration, and spatial normalization to the MNI space [11]. For ADHD-200, we used the fMRIPrep [9, 10] processed data from Bellec et al. [4] and regressed out nuisance variables using cosine bases, six motion parameters, and aCompCor components. Following Kim et al. [17], we set each fMRI volume to the shape of $(96, 96, 96)$ by cropping out the background and padding appropriately, and we then applied global min-max normalization and rescaled it to the $[-1, 1]$ range to match the input range of the DCAE.

We split UKB and HCP using stratified sampling: by age and sex for UKB, and by age, sex, and intelligence score for HCP. For the ADHD-200 dataset, we performed stratified sampling based on diagnosis labels and image acquisition sites, following Kan et al. [15]. We generated four different random stratified splits, and for all of the splits, the training, validation, and test sets were assigned in a $0.7 : 0.15 : 0.15$ ratio. For the ADHD-200 dataset, we experimented with three random training seeds for each split to ensure reliable results, given the relatively small size of the dataset.

Table 1. Performance comparison to baselines on classification and regression tasks. The best results are **bolded** and the second best results are underlined. Standard deviations are provided in Sec. D.

Method	UKB						ADHD-200		
	Sex			Age			Diagnosis		
	ACC	AUC	F1	MSE	MAE	ρ	ACC	AUC	F1
XGBoost	84.1	0.916	0.830	0.698	0.686	0.553	62.3	0.650	0.555
BrainNetCNN	91.7	0.969	0.912	0.597	0.618	0.647	59.2	0.640	0.545
BNT	92.4	0.980	0.919	0.540	0.588	0.685	63.6	0.677	0.624
meanMLP	87.7	0.949	0.919	0.672	0.662	0.586	56.8	0.617	0.532
Brain-JEPA ¹	86.8	0.943	0.862	0.688	0.669	0.574	–	–	–
TFF ($T = 20$)	98.3	<u>0.998</u>	0.982	0.440	0.525	0.760	63.3	0.700	0.608
SwiFT ($T = 20$)	97.4	<u>0.998</u>	0.972	0.366	0.480	0.800	63.3	0.693	0.623
SwiFT ($T = 50$)	<u>98.1</u>	0.999	<u>0.980</u>	<u>0.364</u>	<u>0.477</u>	<u>0.802</u>	<u>63.9</u>	<u>0.701</u>	<u>0.627</u>
TABLET ($T = 256$)	97.7	<u>0.998</u>	0.976	0.340	0.466	0.814	65.8	0.729	0.630

Method	HCP						Intelligence		
	Sex			Age					
	ACC	AUC	F1	MSE	MAE	ρ	MSE	MAE	ρ
XGBoost	82.2	0.890	0.837	0.859	0.769	0.296	0.908	0.779	0.292
BrainNetCNN	86.3	0.937	0.866	0.847	0.749	0.372	0.967	0.788	0.286
BNT	86.3	0.935	0.872	0.794	0.719	0.444	0.920	0.778	0.318
meanMLP	84.5	0.915	0.855	0.846	0.751	0.370	0.887	0.767	0.340
Brain-JEPA	73.9	0.809	0.761	0.814	0.746	0.369	0.959	0.799	0.171
TFF ($T = 20$)	88.1	0.937	0.892	0.888	0.779	0.246	0.898	0.767	0.312
SwiFT ($T = 20$)	<u>93.1</u>	<u>0.978</u>	<u>0.937</u>	0.776	0.719	0.450	0.940	0.782	0.297
SwiFT ($T = 50$)	92.2	0.972	0.929	0.764	0.699	<u>0.460</u>	<u>0.865</u>	<u>0.758</u>	<u>0.354</u>
TABLET ($T = 256$)	93.8	0.987	0.943	<u>0.773</u>	<u>0.705</u>	0.473	0.835	0.741	0.392

Table 2. Performance comparison between TABLET trained from scratch (TFS) and fine-tuned after masked pre-training (FT) on HCP.

Model	HCP								
	ACC	Sex AUC	F1	MSE	Age MAE	ρ	MSE	Intelligence MAE	ρ
TABLET TFS	93.8 $\pm$ 0.9	0.987 $\pm$ 0.003	0.943 $\pm$ 0.008	0.773 $\pm$ 0.077	0.705 $\pm$ 0.038	0.473 $\pm$ 0.053	0.835 $\pm$ 0.053	0.741 $\pm$ 0.028	0.392 $\pm$ 0.062
TABLET FT	95.3 $\pm$ 1.3	<u>0.986</u> $\pm$ 0.005	0.958 $\pm$ 0.011	0.650 $\pm$ 0.045	0.655 $\pm$ 0.024	0.552 $\pm$ 0.032	0.796 $\pm$ 0.051	0.732 $\pm$ 0.028	0.435 $\pm$ 0.046

Prediction Targets and Evaluation Metrics. We considered sex and age for both UKB and HCP, intelligence (CogTotalComp-AgeAdj) for HCP, and diagnosis for ADHD-200. The continuous targets (age, intelligence) are z-normalized using the training set. Classification tasks were evaluated with accuracy, AUC (Area Under ROC Curve), and F1 score. Regression tasks were evaluated with MAE (Mean Absolute Error), MSE (Mean Squared Error), and ρ (Pearson’s correlation).

Baselines. We considered five ROI-based models as our baseline: XGBoost (eXtreme Gradient Boosting) [7] as a standard machine learning baseline, BrainNetCNN [16], Brain Network Transformer (BNT) [15], meanMLP [22], and Brain-JEPA [8]. For the Brain-JEPA, we trained a model from scratch for a fair comparison. To preprocess the data, we first construct the FC matrix using a total of 450 ROIs, comprising 400 ROIs from the Schaefer-400 atlas [28] and 50 additional ROIs from the Tian-Scale III atlas [33]. For the XGBoost model, we used the upper-triangular FC matrix as the input. We followed the preprocessing pipeline of Brain-JEPA for its experiments.

For the voxel-based baselines, we adopted TFF [18] and SwiFT [17], the state-of-the-art voxel-based model. We reproduced the original model with 20 input time frames ($T = 20$) for both of them. We also extended SwiFT to our hardware limit ($T = 50$) to test possible gains from a longer temporal context; alongside the input time frames, the temporal window size was also extended from 4 to 10.

4.2. Main Results

Tab. 1 presents experimental results comparing the performance of different models on a training-from-scratch setting, and the standard deviations are detailed in Sec. D. We also compare the performance between SwiFT and TABLET while matching T in Sec. C.1 to compare the two models in detail.

The results demonstrate that TABLET outperforms baseline methods, including both ROI-based and voxel-based approaches, across four tasks and three datasets, with only

¹ Since the ADHD-200 dataset contains fMRI data with varying repetition time (TR) values and fewer than 160 frames, the default frame number used in Brain-JEPA, we were unable to conduct experiments.

marginal gains on the HCP-Age task and competitive performance on the UKB-Sex task.

Interestingly, the results of SwiFT ($T = 20, 50$) and TABLET indicate a positive association between the number of input time frames (T) and performance, in intelligence prediction and ADHD diagnosis, suggesting that modeling longer temporal variability may be particularly advantageous for these tasks. Sec. 4.6 expands on this observation with a more detailed study.

Although the overall improvement over SwiFT appears modest, we believe this is likely due to two factors: first, the nature of rs-fMRI, in which signals are recorded while subjects are at rest; and second, the tasks considered in our study, which primarily rely on structural information and may not benefit from longer temporal dependencies.

We expect that applying TABLET to task fMRI data, which contain more temporally dynamic signals, would result in more substantial performance gains over the baselines as it can cover much longer temporal sequences. To support this, we provide preliminary experimental results on HBN-Movie [2] in Sec. C.2, where each subject watches two different movies during fMRI acquisition. The results suggest that TABLET benefits more from extended temporal context compared to the baselines, demonstrating its potential advantage in settings with highly dynamic stimuli. In addition, investigating a broader range of tasks to determine which ones can benefit from longer temporal information may provide valuable scientific insights. We believe these are promising directions for future work.

4.3. Effect of Pre-training on Downstream Tasks

Tab. 2 shows the effectiveness of the masked token pre-training strategy described in Sec. 3.3. We first pre-trained TABLET on a large UKB dataset with a 9:1 training and validation split. We then fine-tuned the model on HCP to simulate a transfer learning setting. For fine-tuning, we only used 10 epochs, which is considerably fewer compared to training from scratch. The results demonstrate that the pre-training of TABLET indeed contributes to the improvement of downstream task performance, albeit with varying amounts of success depending on the task.

4.4. Comparison of 2D Natural Image DCAE and 3D fMRI-trained DCAE

To compare the quality of the tokens generated by 2D DCAE and 3D DCAE, we evaluate them in two different aspects: reconstruction quality and downstream performance. The reconstruction quality would serve as a proxy of the crucial information preserved in the latent representations, while the downstream performance would represent the practical impact of the two.

Reconstruction quality. We first compare the quality of the fMRI reconstructions from the two autoencoders.

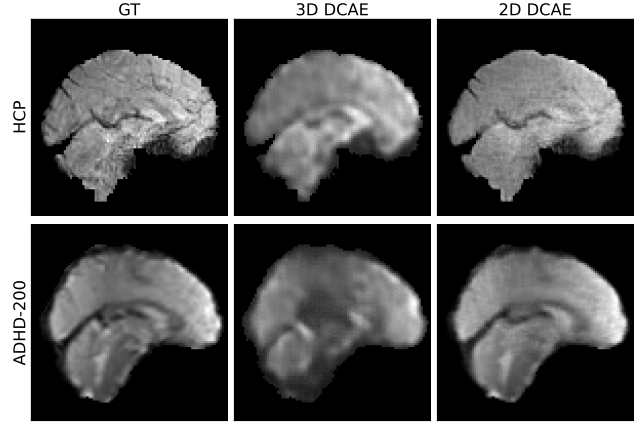


Figure 3. Visualization of reconstructions from 3D and 2D DCAE.

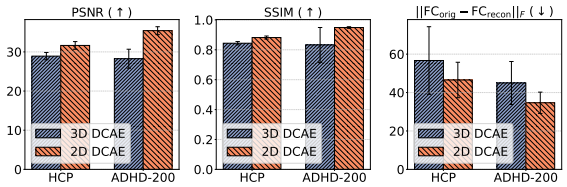


Figure 4. Information preservation of 3D and 2D DCAE.

The reconstructions were assessed in two different levels: first, we quantified voxel-level reconstruction fidelity using PSNR, which reflects the preservation of fine-grained intensity details, and SSIM, which captures structural similarity and local spatial patterns. Second, we evaluated the preservation of high-level functional information by comparing the difference in FC matrix between the original fMRI data (FC_{orig}) and its reconstruction (FC_{recon}): $\|FC_{orig} - FC_{recon}\|_F$.

The 3D DCAE was trained with the UKB dataset; a detailed training procedure is provided in Sec. B. To assess generalizability, we deliberately excluded HCP and ADHD-200 from the training set. The reconstructions from the three slicing axes were averaged for 2D DCAE.

The 2D DCAE moderately outperformed the 3D DCAE both qualitatively and quantitatively, as shown in Fig. 3 and Fig. 4. This result suggests that the 2D DCAE, pre-trained on massive, diverse natural image datasets, learns highly robust and general-purpose low-level feature extractors (e.g., for edges, textures), and in turn is able to generalize well into medical images. Moreover, the 2D DCAE also better preserved functional connectivity patterns, indicating that its latent representations retain high-level functional relationships in the brain.

As a side note, we also attempted to fine-tune the 2D DCAE with fMRI data while freezing different parts of the autoencoder, but discovered that any fine-tuning consistently harmed the reconstruction quality. We presume this is because our fMRI dataset is relatively small and homo-

Table 3. Performance comparison between TABLET with latents from 3D DCAE and 2D DCAE on HCP and ADHD-200. Standard deviations are provided in Sec. D.

Tokenizer	HCP									ADHD-200		
	Sex			Age			Intelligence			Diagnosis		
	ACC	AUC	F1	MSE	MAE	ρ	MSE	MAE	ρ	ACC	AUC	F1
3D DCAE	92.2	0.973	0.929	0.767	0.693	0.475	0.869	0.755	0.387	65.8	0.711	0.644
2D DCAE	93.8	0.987	0.943	0.773	0.705	0.473	0.835	0.741	0.392	65.8	0.729	0.630

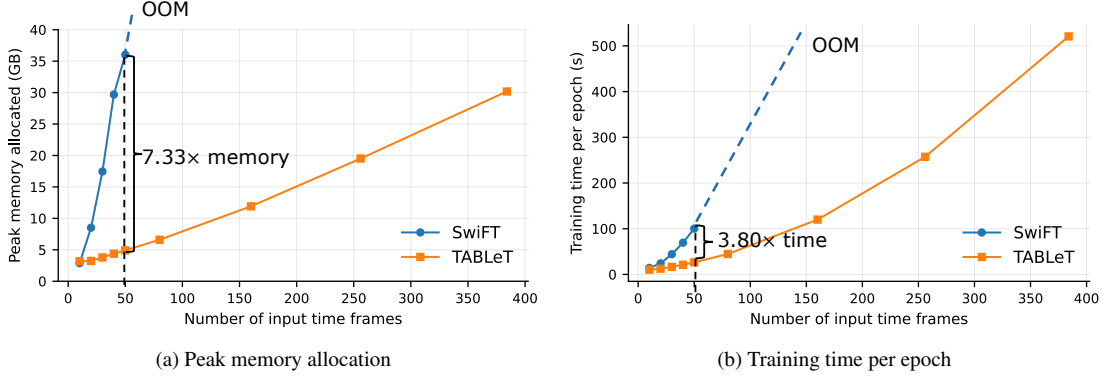


Figure 5. Comparison of (a) memory and (b) training time, between TABLET and SwiFT.

geneous compared to the dataset the model is trained on, potentially harming generic filters crucial for the model’s generalization capabilities.

Downstream Performance. We also compare TABLET models trained with latents from the 3D DCAE and the 2D DCAE. The experiment setup is identical to the one detailed in Sec. 4.1 and Sec. 4.2. As shown in Tab. 3, both models achieve comparable performance, with the 2D DCAE moderately outperforming the 3D counterpart in most cases.

Overall, these results indicate that a 2D DCAE, despite never being trained on fMRI, generalizes well as a tokenizer for 4D fMRI, both in terms of reconstruction fidelity and downstream performance. Given its strong performance, alongside the practical benefit of avoiding costly dataset collection or additional fine-tuning, we advocate using an existing 2D DCAE as a training-free tokenizer rather than building and training a dedicated 3D counterpart.

4.5. Memory and Computational Efficiency

We conduct a quantitative analysis to compare the memory and computational efficiency between TABLET and SwiFT. To ensure a fair comparison, all tests were performed on a single NVIDIA RTX A6000 GPU, and the batch size of both models was fixed to 4. We were only able to run SwiFT up to $T = 50$ due to memory limitations. At $T = 50$, compared to SwiFT, TABLET is 7.33 times more memory efficient, and trains 3.8 times faster. With a similar memory budget (~ 30 GB), T can be extended nearly tenfold between SwiFT ($T = 40$) and TABLET ($T = 384$).

4.6. Additional Ablation Studies

Effect of Aggregation of Three Axes. We examined the effect of axis aggregation to better understand its effect: we compared models trained with fMRI tokens derived from a single axis alongside the model with aggregated tokens. As shown in Tab. 4, the performance of TABLET varies depending on the chosen axis for single-axis models. In contrast, our aggregated version consistently achieves strong performance across tasks, eliminating the dependence on any particular slicing axis. We chose to aggregate all three axes instead of using a single axis due to this reliability.

Effect of T . As shown in Tab. 1, modeling longer-range

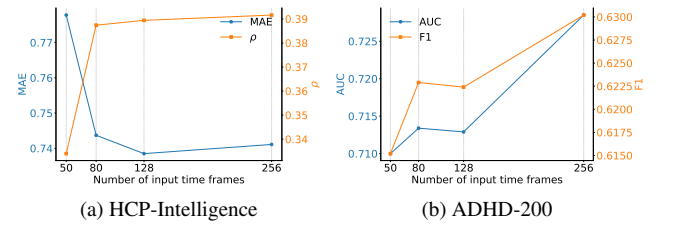


Figure 6. Performance of TABLET on HCP-Intelligence and ADHD-200 with varying T .

temporal dynamics can improve performance on the HCP-Intelligence and ADHD diagnosis tasks. To explore this further, we varied the T of TABLET and evaluated the corresponding performance. Interestingly, Fig. 6 reveals a clear positive trend between performance and T . We believe that investigating the relationship between T and model performance across diverse tasks represents a promising direction

Table 4. Effect of the choice of slicing axis and aggregation of the three axes on classification and regression tasks. The best results are **bolded** and the second best results are underlined.

Axis	UKB					
	Sex			Age		
	ACC	AUC	F1	MSE	MAE	ρ
Sagittal	<u>97.3</u> ± 0.9	0.996 ± 0.001	<u>0.971</u> ± 0.009	0.369 ± 0.015	<u>0.486</u> ± 0.010	<u>0.796</u> ± 0.008
Coronal	97.1 ± 0.4	0.996 ± 0.002	0.969 ± 0.004	0.435 ± 0.012	0.525 ± 0.008	0.756 ± 0.009
Axial	<u>97.3</u> ± 0.4	<u>0.997</u> ± 0.000	<u>0.971</u> ± 0.004	0.410 ± 0.020	0.509 ± 0.014	0.771 ± 0.013
All	97.7 ± 0.2	0.998 ± 0.000	0.976 ± 0.002	0.340 ± 0.011	0.466 ± 0.010	0.814 ± 0.009

Axis	HCP					
	Sex			Age		
	ACC	AUC	F1	MSE	MAE	ρ
Sagittal	91.3 ± 3.6	0.972 ± 0.017	0.920 ± 0.033	0.783 ± 0.111	0.721 ± 0.041	0.458 ± 0.076
Coronal	<u>93.6</u> ± 1.7	<u>0.981</u> ± 0.007	<u>0.941</u> ± 0.015	0.855 ± 0.053	0.745 ± 0.023	0.376 ± 0.048
Axial	92.3 ± 3.0	0.979 ± 0.008	0.930 ± 0.028	0.748 ± 0.056	<u>0.711</u> ± 0.015	<u>0.470</u> ± 0.040
All	93.8 ± 0.9	0.987 ± 0.003	0.943 ± 0.008	<u>0.773</u> ± 0.077	0.705 ± 0.038	0.473 ± 0.053

Axis	HCP			ADHD-200		
	Intelligence			Diagnosis		
	MSE	MAE	ρ	ACC	AUC	F1
Sagittal	<u>0.842</u> ± 0.058	<u>0.744</u> ± 0.028	0.401 ± 0.060	65.8 ± 2.3	<u>0.715</u> ± 0.026	0.633 ± 0.032
Coronal	0.850 ± 0.057	0.749 ± 0.029	0.381 ± 0.065	63.5 ± 3.1	0.707 ± 0.036	0.621 ± 0.040
Axial	0.896 ± 0.070	<u>0.773</u> ± 0.033	0.309 ± 0.072	<u>64.3</u> ± 2.5	0.713 ± 0.022	0.622 ± 0.034
All	0.835 ± 0.053	0.741 ± 0.028	<u>0.392</u> ± 0.062	65.8 ± 3.5	0.728 ± 0.020	<u>0.630</u> ± 0.038

for future research.

4.7. Interpretation Results

One advantage of voxel-based methods is that the models are interpretable, since the entire process from voxel to prediction is differentiable. To test the interpretability of TABLET, we used Integrated Gradients (IG) [32] for visualization of highly contributing areas for sex-classification. We used female test subjects in the HCP-Sex task who are correctly classified with TABLET with high confidence ($\geq 75\%$), and computed the IG map of the first frame from each subject, then averaged it.

Fig. 7 shows that TABLET mainly focuses on the medial prefrontal gyrus (mPFC), posterior cingulate cortex (PCC), precuneus (PCu), and thalamus (Thal.), regions commonly implicated in brain sex difference literature [12, 26, 27, 36].

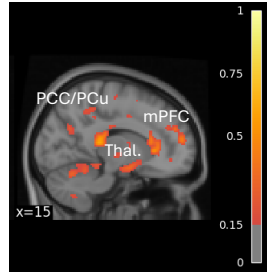


Figure 7. IG map.

5. Conclusion & Limitations

We introduced TABLET, a simple and scalable framework that uses a natural-image 2D autoencoder as a training-free tokenizer for fMRI volumes, enabling long-range temporal modeling with a lightweight Transformer. Across three datasets, TABLET achieved competitive or better perfor-

mance while offering substantial reductions in memory and computation compared to voxel-based baselines. In addition, we show that masked token pre-training further enhances downstream performance. Our model is based on a striking result: a 2D autoencoder trained on natural images preserves fMRI structure and functional relationships better than a 3D fMRI-trained autoencoder, enabling efficient tokenization without any domain-specific training. Our model opens the opportunity to handle much longer temporal sequences for future researchers.

Despite these advantages, our study has several limitations. First, TABLET tokenizes each frame of the fMRI time series independently. This process may disrupt subtle temporal dynamics. Tokenization strategies that directly incorporate temporal dependencies, especially in tasks where fine-grained dynamics are critical, could be explored in the future. Second, TABLET processes all tokens jointly, without explicit modeling of their spatial or temporal structure. Architectures designed to leverage spatial and temporal alignment between tokens may further enhance the ability to capture the spatiotemporal dynamics inherent in fMRI data. Moreover, as discussed in Sec. 4.2, the benefits of longer temporal modeling vary across tasks, and a systematic understanding of which neuroscientific phenomena gain the most remains open.

Nevertheless, we believe our study suggests a promising approach, bridging natural image processing and medical imaging, and enabling scalable, efficient spatiotemporal modeling of brain activity.

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